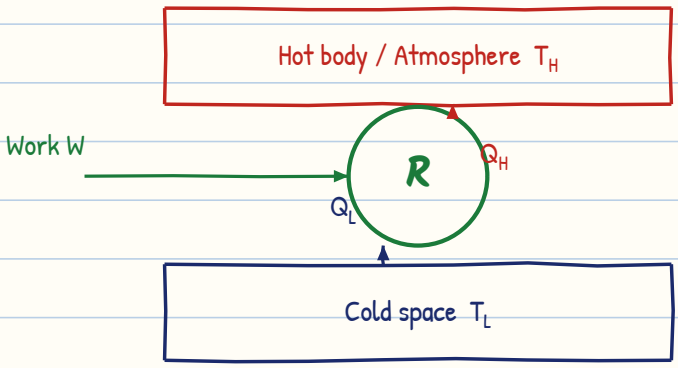
Syllabus: Necessity and applications, Carnot refrigerator, first and second law applied to refrigerating machines, unit of refrigeration, COP, EER, different refrigeration methods. Air refrigeration: Bell-Coleman cycle, ideal and actual cycles, open and dense air systems, numericals, refrigeration needs of aircraft.

I. Introduction to Refrigeration

- **Refrigeration** = the process of removing heat from a space or substance so that its temperature is lowered below that of the surroundings, and then keeping it there.
- It needs a device (refrigerating machine) which takes in **work** (or heat) and pumps heat from a low-temperature body to a high-temperature body. This is not natural: heat flows hot → cold by itself. **Second law** says we must spend work.
- **Refrigerator** → aim: extract Q_L from the cold space. **Heat pump** → aim: deliver Q_H to the warm space. Same machine, different purpose.



$Q_H = Q_L + W$

$COP = Q_L / W$

Necessity of refrigeration

- Preservation of food (milk, meat, fish, fruits, vegetables) – slows bacterial growth and enzyme action, so shelf life increases.
- Comfort for human beings: air conditioning of homes, offices, theatres, hospitals, vehicles.
- Industrial process control: machine tools, precision measuring rooms, textile (humidity control), printing, photography film.
- Storage of medicines, vaccines, blood banks (cold chain).

Applications

- **Domestic:** refrigerator, deep freezer, water cooler, room AC.

- **Commercial:** cold storage, ice plant, hotels, supermarkets, display cabinets.
- **Industrial:** chemical industry (separation of gases, liquefaction of O_2, N_2, LPG), petroleum dewaxing, steel (cooling of blast furnace air), dairy, brewery.
- **Transport:** refrigerated trucks, rail vans, ships, aircraft cabin cooling.
- **Others:** cooling concrete of dams, mining (cooling mines), ice skating rinks, cryogenics, space research, superconductors.

2. First and Second Law applied to Refrigerating Machines

First law (energy conservation)

For a cyclic device, net heat = net work. Taking the refrigerator as the system (heat out is +ve):

$$Q_H = Q_L + W$$

where Q_L = heat absorbed from cold space, Q_H = heat rejected to hot body, W = work input.

Second law

- **Clausius statement:** It is impossible for a self-acting machine, unaided by external agency, to move heat from a body at lower temperature to one at higher temperature.
- **Kelvin-Planck statement:** It is impossible to construct a device operating in a cycle that produces net work while exchanging heat with a single reservoir.
- Meaning for refrigerator: W cannot be zero, hence COP is always finite. A refrigerator with $W = 0$ (a "perpetual motion machine of second kind") is impossible.
- The two statements are equivalent: violation of one means violation of the other.

Coefficient of Performance (COP)

$$COP_R = \text{Refrigerating effect} / \text{Work input} = Q_L / W = Q_L / (Q_H - Q_L)$$

$$COP_{HP} = Q_H / W = Q_H / (Q_H - Q_L) \; ; \; COP_{HP} = COP_R + 1$$

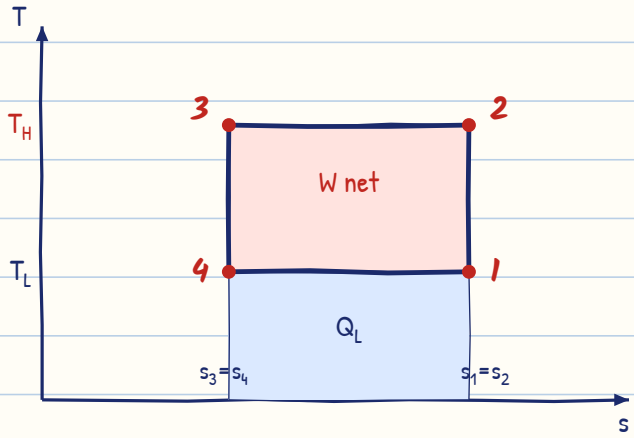
Note: COP of a refrigerator can be greater than 1 (it is not an efficiency, since it is output/input of different forms of energy). COP of heat pump is always more than 1.

Heat pump (reversed cycle) – quick idea

- The same cycle used to heat a space: Q_H is the useful effect. $COP_{HP} = Q_H/W = COP_R + 1 > 1$.
- Example: a refrigerator of COP 3 used as heat pump gives COP 4, so 1 kW of electricity delivers 4 kW of heat – much better than a 1 kW heater.
- Carnot heat pump between 5 °C and 35 °C: COP = 308/30 = **10.27**.

3. Carnot Refrigerator (Reversed Carnot Cycle)

The Carnot cycle is the most efficient reversible cycle between two temperatures T_H and T_L. Reversing the Carnot engine gives the Carnot refrigerator.



1-2 Isentropic compression ($T_L \rightarrow T_H$)

2-3 Isothermal heat rejection at T_H (Q_H)

3-4 Isentropic expansion ($T_H \rightarrow T_L$)

4-1 Isothermal heat absorption at T_L (Q_L)

Area 4-1-s-s = Q_L, Area 1-2-3-4 = W

- Heat absorbed (4-1): $Q_L = T_L (s_1 - s_4)$
- Heat rejected (2-3): $Q_H = T_H (s_2 - s_3)$ with $s_1=s_2$, $s_3=s_4$
- Net work: $W = Q_H - Q_L = (T_H - T_L)(s_1 - s_4)$

$$COP_{Carnot} = T_L / (T_H - T_L) \; ; \; COP_{HP,Carnot} = T_H / (T_H - T_L)$$

Observations (important for exams)

- COP depends **only on temperatures** of the reservoirs, not on the working fluid.
- COP rises when T_L is raised or T_H is lowered (i.e. smaller temperature lift).
- No actual refrigerator can beat Carnot COP between the same temperatures. It is the **upper limit** and the yard-stick for comparison. **Relative COP = Actual COP / Carnot COP**

Why Carnot cycle is not used practically

- Isothermal heat transfer needs infinitely slow processes; compression of a wet vapour (state 1) is difficult – liquid droplets damage the compressor (wet compression).

- Isentropic expansion 3-4 (a work-recovering expander) of liquid-vapour mixture gives little work – not worth the cost; replaced by a throttle valve in VCR.
- For gases (air) isothermal processes are impossible in a machine → Bell-Coleman cycle.

4. Unit of Refrigeration

- The capacity of a refrigerating machine is measured in **Tonne of Refrigeration (TR)**.
- 1 TR = the heat to be removed from 1 tonne (2000 lb, short ton) of water at 0 °C to form ice at 0 °C in 24 hours.

$$1\ TR = 2000 \times 144\ BTU / 24\ h = 12\ 000\ BTU/h = 200\ BTU/min$$

$$1\ TR = 211\ kJ/min = 3.5167\ kW \approx 3.5\ kW \approx 3024\ kcal/h$$

- Latent heat of ice = 335 kJ/kg (144 BTU/lb). 1 BTU = 1.055 kJ.
- Other units: 1 kW = 3412 BTU/h = 860 kcal/h. Domestic fridge ≈ 0.1 TR; window AC 1-2 TR.

EER and other performance indices

- **EER** (Energy Efficiency Ratio) = cooling capacity in BTU/h divided by electrical power input in Watts.

$$EER = Q_L(BTU/h) / P(W)\ ;\ EER = 3.412 \times COP$$

- **SEER** = seasonal EER, the cooling delivered over a whole season / energy consumed (more realistic). **ISEER** in India.
- Another measure: kW/TR = 3.5167 / COP (lower is better). For chillers a good value is 0.6 kW/TR.

Index	Unit	Typical value
COP	dimensionless	2.5 - 5 (air-conditioner)
EER	BTU/h per W	9 - 12
kW/TR	kW per TR	0.6 - 1.2
SEER	BTU/h per W	13 - 22

5. Different Methods of Refrigeration

Refrigeration methods are classified into **non-cyclic** (one-time cooling) and **cyclic** (continuous).

Non-cyclic methods

- **Ice refrigeration:** ice melts at 0 °C absorbing 335 kJ/kg; used in fish and vegetable transport. Cheap but temperature limited to ≈ 0 °C.
- **Ice + salt (freezing mixtures):** gives temperature below 0 °C (NaCl: −21 °C, $CaCl_2$: −50 °C).
- **Dry ice (solid CO_2):** sublimes at −78.5 °C at 1 atm, latent heat 573 kJ/kg; used for ice cream and transport of frozen goods.
- **Evaporative cooling:** water evaporates and absorbs latent heat from air (desert cooler); works only in dry climate.

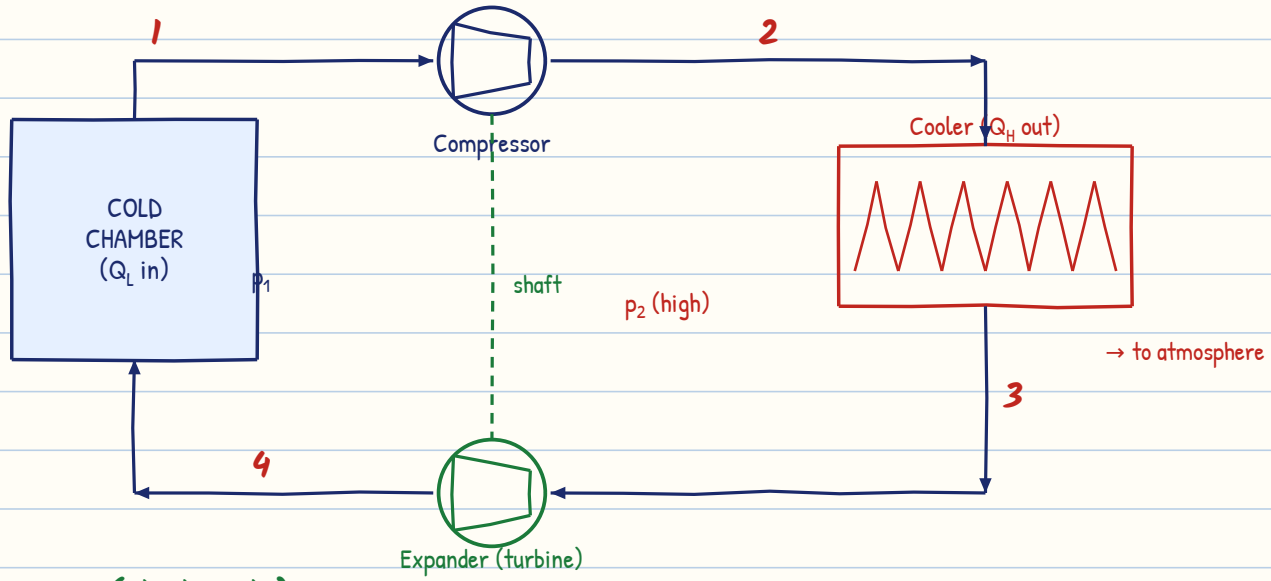
Cyclic methods

Method	Working fluid	Energy input
Air refrigeration (Bell-Coleman)	Air	Mechanical work
Vapour compression (VCR)	Freon / NH_3	Mechanical work
Vapour absorption (VAR)	NH_3-water, LiBr-water	Heat
Steam jet	Water vapour	High pressure steam
Thermo-electric (Peltier)	Electrons in junction	DC electricity
Vortex tube (Hilsch)	Compressed air	Pressure energy
Magnetic (adiabatic demag.)	Paramagnetic salt	Magnetic field

- Cryogenic (gas liquefaction – Linde, Claude) methods reach below −150 °C.

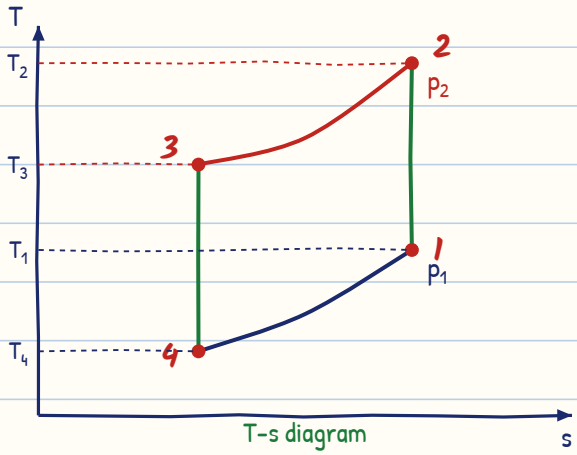
6. Air Refrigeration – Bell-Coleman (Reversed Joule / Brayton) Cycle

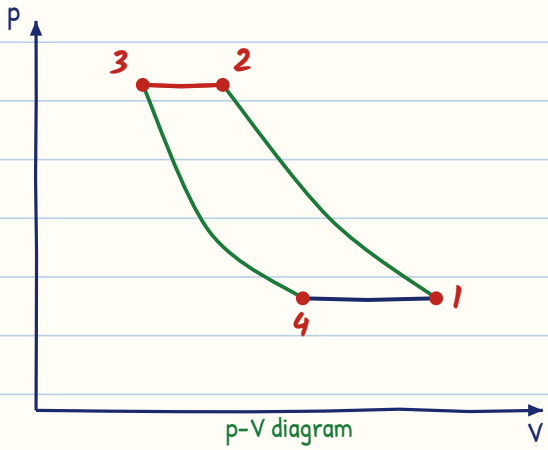
- Working fluid is **air** (a gas, no phase change). Also called the **reversed Brayton** or **Joule cycle**. Used in aircraft cooling and gas liquefaction.
- Heat rejection and absorption happen at **constant pressure**, not constant temperature (a gas cannot absorb heat isothermally in a heat exchanger).



Processes (ideal cycle)

- **1-2** Isentropic compression of air from cold chamber (p_1, T_1) to (p_2, T_2) – work input.
- **2-3** Constant pressure cooling in cooler – heat Q_H rejected to the atmosphere, $T_3 > T_0$.
- **3-4** Isentropic expansion in expander (turbine) to p_1; temperature falls to T_4 T_1; work is recovered.
- **4-1** Constant pressure heating in cold chamber – heat Q_L absorbed from the refrigerated space.





Cycle analysis

Take unit mass of air, c_p = 1.005 kJ/kg K, γ = 1.4.

$$Q_L = c_p(T_1 - T_4) \quad Q_H = c_p(T_2 - T_3)$$

$$W_{net} = W_c - W_e = c_p[(T_2 - T_1) - (T_3 - T_4)]$$

$$COP = (T_1 - T_4) / [(T_2 - T_1) - (T_3 - T_4)]$$

For isentropic processes 1-2 and 3-4: $T_2/T_1 = T_3/T_4 = (p_2/p_1)^{(\gamma-1)/\gamma} = r_p^{(\gamma-1)/\gamma}$

$$COP = 1 / [\, r_p^{(\gamma-1)/\gamma} - 1 \,] = T_4 / (T_3 - T_4) = T_1 / (T_2 - T_1)$$

Remarks

- COP **decreases** with increase of pressure ratio r_p. (but a larger r_p gives lower T_4, more cooling per kg).
- COP of Bell-Coleman < Carnot COP between T_1 and T_3 (because the cycle works over a wider temperature range T_4 to T_2).
- Refrigerating effect is small (cp of air low) → large mass flow, bulky machine. Hence low COP, used only where air is needed (aircraft).

Merits and demerits of air refrigeration

- **Merits:** air is cheap, non-toxic, non-flammable, no leakage problem; light and simple for aircraft; can give very low temperatures (−60 °C or less) by increasing r_p.
- **Demerits:** low COP (≈ 0.5 - 2); large mass flow, bulky equipment; moisture in air freezes and chokes the expander; higher running cost; power needed is high.

Item	Air cycle	Vapour compression
Working fluid	Air (gas)	Refrigerant (changes phase)
Heat transfer	Constant pressure, sensible	Constant T & p, latent
COP	Low (≈ 1)	High (3 - 5)
Equipment size	Large	Compact
Main use	Aircraft, liquefaction	Homes, industry, cold storage

Actual Bell-Coleman cycle

- Real compression is not isentropic: state 2 → **2'** (more work). Real expansion gives 4 → **4'** (higher T, less cooling).
- Pressure drops in coolers, pipes and cold chamber; heat leakage.

$$\eta_c = (T_2 - T_1) / (T_2' - T_1) \quad \eta_e = (T_3 - T_4') / (T_3 - T_4)$$

$$COP_{actual} = (T_1 - T_4') / [(T_2' - T_1) - (T_3 - T_4')]$$

Regenerative air cooling (gas refrigeration with regenerator)

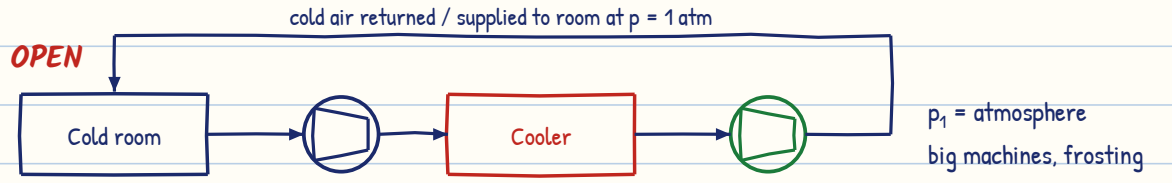
- Warm air leaving the cooler (state 3) is further cooled in a **regenerator** by the cold air coming from the cold chamber, before entering the expander. The expander inlet becomes $T_3' < T_3$, so lower exit temperature T_4 is reached with the **same pressure ratio**.
- Cold air leaving the cold chamber is pre-heated in the regenerator before compression: compressor work rises a little, but the refrigeration effect rises much more.
- Used in air liquefaction (Linde-Hampson, Claude) and in supersonic aircraft. Energy balance of regenerator: $T_3 - T_3' = T_1' - T_1$.

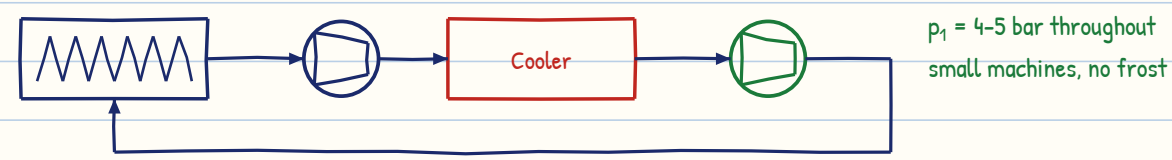
$$Q_L = c_p(T_1 - T_4) \;,\; W = c_p[(T_2 - T_1') - (T_3' - T_4)]$$

Points to remember while solving air-cycle numericals

- Always convert temperatures to Kelvin and pressures to the same unit (bar or kPa).
- Find T_2 and T_4 using the isentropic relation first, then correct with η_c and η_e.
- Refrigerating effect = $c_p(T_1 - T_4)$ per kg; mass flow = capacity (kW)/effect.
- Power = $m \times W_{net}$; COP = capacity / power. Compare with Carnot COP to show relative COP.

Open and Dense air systems





Open air system	Dense (closed) air system
Air from cold chamber is compressed, cooled, expanded and then **discharged to the cold chamber** (or fresh air from atmosphere is cooled and supplied)	Air circulates in a **closed loop** at a pressure above atmospheric throughout; the cold chamber is cooled through a heat exchanger
Lower pressure at 1 = atmospheric, so large volume of air → big machines	Higher density (hence "dense") → smaller compressor & expander for the same capacity
Moisture in air freezes in the expander → choking	Air can be dried; no frosting problem
Low COP	Higher COP due to lower pressure ratio at same pressure level

Note: Dense air: pressure ratio is the same but the low-pressure side is kept at, say, 4-5 bar instead of 1 bar, so the pressure difference (and friction/leakage) are lower and the **size reduces**.

7. Numerical Problems

Problem 1 (ideal cycle)

A Bell-Coleman refrigerator takes air at 1 bar and 270 K (cold chamber), compresses it to 5 bar; after cooling the air leaves the cooler at 300 K. Find: (i) T_2, T_4 (ii) COP (iii) mass flow and power for 10 TR. ($\gamma = 1.4$, $c_p = 1.005$) **Solution:**

$$T_2 = 270 \times 5^{0.2857} = 427.6 \text{ K } T_4 = 300 / 5^{0.2857} = 189.4 \text{ K}$$

$$Q_L = 1.005\ (270 - 189.4) = 81.0 \text{ kJ/kg } W = 1.005\ [(427.6-270) - (300-189.4)] = 47.3 \text{ kJ/kg}$$

$$COP = 81.0 / 47.3 = 1.71$$

Capacity = 10 × 3.5167 = 35.17 kW → mass flow m = 35.17/81.0 = **0.434 kg/s** (26 kg/min); Power = m W = 0.434 × 47.3 = **20.5 kW**

Note: Carnot COP between 270 K and 300 K = 270/30 = 9, much higher than 1.71.

Problem 2 (Carnot)

A Carnot refrigerator works between −10 °C and 30 °C. Find COP and power needed for 10 TR.

$$COP = 263 / (303 - 263) = 6.575 \; P = 35.167 / 6.575 = \mathbf{5.35\ kW}$$

Problem 3 (actual cycle)

Air at 263 K, 1 bar is compressed to 4 bar; cooled to 298 K, expanded back to 1 bar. η_c = 85 %, η_e = 90 %. Find COP.

$$T_2 = 263 \times 4^{0.2857} = 390.8\ K \; ; \; T_4 = 298/4^{0.2857} = 200.5\ K$$

$$T_2' = 263 + (390.8-263)/0.85 = 413.4\ K \; ; \; T_4' = 298 - 0.9(298 - 200.5) = 210.3\ K$$

$$Q_L = 1.005\ (263 - 210.3) = 53.0\ kJ/kg \; ; \; W = 1.005\ [(413.4-263) - (298-210.3)] = 63.0\ kJ/kg$$

$$COP_{actual} = 53.0 / 63.0 = \mathbf{0.84}\ (ideal = 1/(4^{0.2857}-1) = 1.97)$$

Problem 4 (unit conversion)

A cold store needs 100 kW cooling. Express in TR: 100 / 3.5167 = **28.4 TR**. If EER is 10 → electrical power = (100×3412/1000 BTU/h = 341 200 BTU/h)/10 = 34.1 kW.

Problem 5 (aircraft simple system)

An aircraft flies at 250 m/s where ambient air is 0.7 bar, 250 K. Ram air is brought to rest isentropically. Bleed air (ram state) is compressed to 4.5 bar (η_c = 80 %), cooled to the ram temperature + 20 K, expanded in a turbine to cabin pressure 1.013 bar (η_t = 85 %). Cabin is held at 298 K and load is 20 kW. Find air mass flow and COP. **Solution:**

$$T_0 = 250 + 250^2/(2\times1005) = 281.1\ K \; ; \; p_0 = 0.7\ (281.1/250)^{3.5} = 1.055\ bar$$

$$T_2 = 281.1\ (4.5/1.055)^{0.2857} = 425.4\ K \; ; \; T_2' = 281.1 + (425.4-281.1)/0.8 = 461.5\ K$$

$$T_3 = 281.1 + 20 = 301.1\ K \; ; \; T_4 = 301.1\ (1.013/4.5)^{0.2857} = 196.6\ K \; ; \; T_4' = 301.1 - 0.85\ (301.1-196.6) = 212.3\ K$$

$$m = 20 / [1.005 (298 - 212.3)] = \mathbf{0.232\ kg/s}$$

$$W_c = m\, c_p(T_2' - T_0) = 42.1\ kW\ ;\ W_t = m\, c_p(T_3 - T_4') = 20.7\ kW\ ;\ W_{net} = 21.4\ kW$$

$$COP = 20 / 21.4 = \mathbf{0.935}$$

Problem 6 (effect of pressure ratio)

For $T_1 = 270$ K, $T_3 = 300$ K, find COP for $r_p = 3$ and 5.

$$r_p = 3:\ COP = 1/(3^{0.2857} - 1) = \mathbf{2.71}\ (T_4 = 219.2\ K)$$

$$r_p = 5:\ COP = 1/(5^{0.2857} - 1) = \mathbf{1.71}\ (T_4 = 189.4\ K)$$

Note: Higher pressure ratio → colder air but lower COP. Designer must compromise.

Problem 7 (ice making)

How much power is needed to make 5 tonnes (5000 kg) of ice per day from water at 25 °C to ice at −5 °C, if the actual COP is 3.5? (c_{pw}=4.187, L=335, c_{pi}=2.1 kJ/kg K)

Heat removed per kg = 4.187×25 + 335 + 2.1×5 = 104.7 + 335 + 10.5 = 450.2 kJ/kg

$$Q_L = 5000 \times 450.2 / (24 \times 3600) = 26.05\ kW = \mathbf{7.41\ TR}$$

Power = 26.05 / 3.5 = **7.44 kW**

Problem 8 (EER)

A 1.5 TR air conditioner draws 1.6 kW. Find COP and EER.

$$Q_L = 1.5 \times 3.5167 = 5.275\ kW\ ;\ COP = 5.275/1.6 = \mathbf{3.30}\ ;\ EER = 3.412 \times 3.30 = \mathbf{11.25}$$

8. Refrigeration Needs of Aircraft

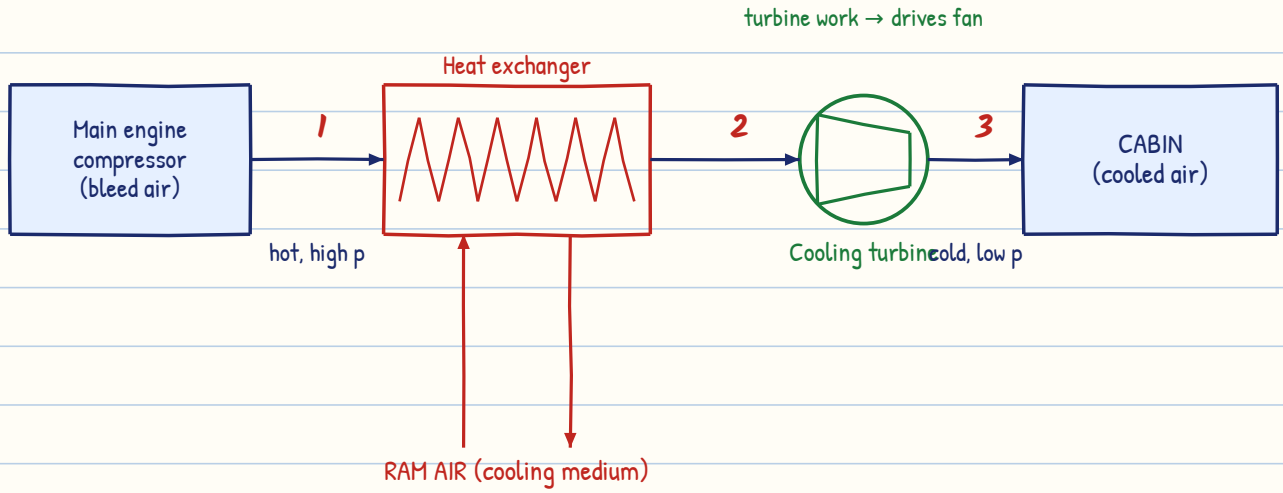
Why air conditioning of aircraft is needed

- **Solar radiation** through windows and skin.

- **Skin friction (kinetic heating)** – at high speed the air stagnates on the skin and heats it. Stagnation temperature rise: $\Delta T = V^2/(2c_p) = T_a(\gamma-1)M^2/2$. At Mach 2 it is more than 100 K.
- **Metabolic heat** of passengers and crew (~100 W each), engine heat, hot electronic/avionic equipment and lighting.
- At altitude, ambient pressure and temperature are low (−50 °C at 10 km) so cabin must be **pressurised** and kept warm in cruise but cooled on ground and at high Mach.

Why air cycle is used in aircraft

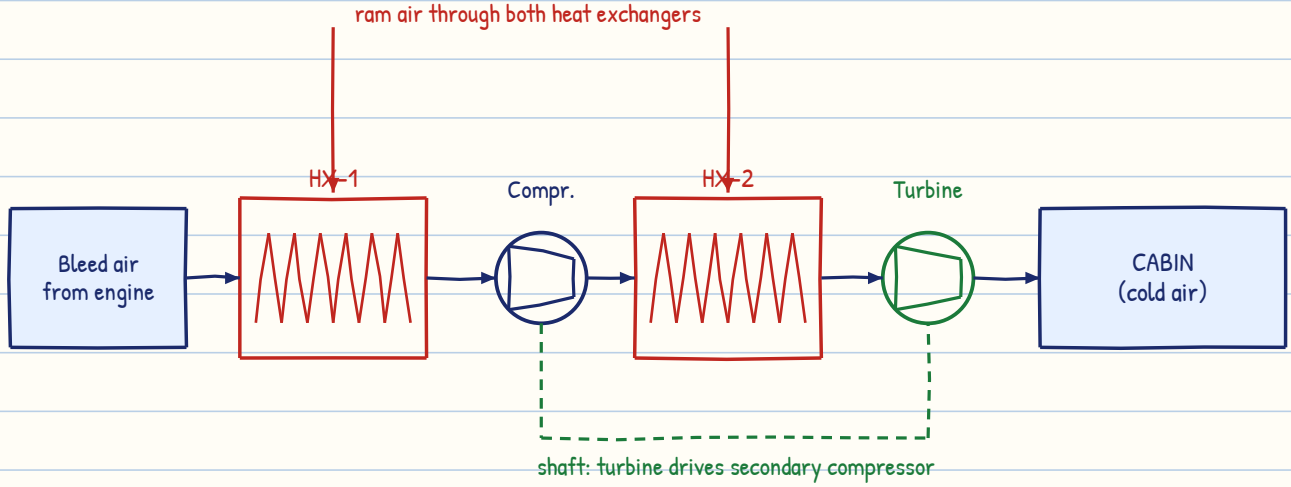
- Air is **freely available** (bleed air from engine compressor / ram air) – no refrigerant storage or leakage hazard.
- **Light weight and compact** system; small cooling turbine; high-pressure air available for free.
- Safe (non-toxic, non-flammable); simple maintenance; works at high altitude where vapour systems fail.
- Low COP does not matter, as the energy is a small part of engine power.



Types of aircraft air-cooling systems

- **Simple system:** bleed air → heat exchanger (ram air cooled) → cooling turbine → cabin (shown above).
- **Boot-strap system:** an additional compressor driven by the turbine raises pressure before the second heat exchanger → lower turbine exit temperature, used for high speed flight.

- **Regenerative system:** a regenerator cools the air before the turbine using the cold turbine exit air; for supersonic aircraft.
- **Reduced ambient system:** cooling turbine outlet is at ambient pressure reduced to altitude pressure; best for very high Mach numbers.



Ram effect

$$T_0 = T_a + V^2/(2\,c_p) = T_a[1 + (\gamma-1)M^2/2]$$

Example: $M = 0.8$, $T_a = 250$ K $\rightarrow T_0 = 250\,(1 + 0.2\times0.64) = 282$ K (rise of 32 K). Ram efficiency $\eta_r = (p_0 - p_a)/(p_0' - p_a)$ accounts for losses in the intake.

9. Quick Revision and Important Questions

- $COP_{Carnot} = T_L/(T_H - T_L)$; 1 TR = 3.5167 kW = 211 kJ/min; EER = 3.412 COP.
- Bell-Coleman COP = $1 / (r_p^{(\gamma-1)/\gamma} - 1)$; actual cycle includes η_c, η_e.
- Heat pump COP = refrigerator COP + 1.

One-page summary (mind map)

Topic	Key formula / idea
Reversed Carnot	$COP = T_L/(T_H-T_L)$; upper limit
TR	1 TR = 3.5167 kW = 211 kJ/min = 12000 BTU/h
Heat pump	$COP_{HP} = COP_R + 1$
Bell-Coleman	$COP = 1/(r_p^{0.2857} - 1)$; 4 processes: 2 isentropic + 2 constant p
Actual cycle	$\eta_c = (T_2-T_1)/(T_2'-T_1)$; $\eta_e = (T_3-T_4')/(T_3-T_4)$
Dense air	Closed loop, p_1 > atm, smaller size
Aircraft	Ram effect, bleed air, simple / boot-strap / regenerative / reduced ambient

Short answers (2 marks)

- **Define 1 TR.** Heat removal rate to freeze 1 tonne of water at 0 °C into ice at 0 °C in 24 h = 3.5167 kW = 211 kJ/min.
- **Why is COP of a refrigerator > 1?** Because it moves heat Q_L using little work W; the heat is not converted from work.
- **What is relative COP?** Ratio of actual COP to theoretical (Carnot) COP; always less than 1.
- **What is dense air system?** Closed air cycle with pressure above atmosphere on low side – smaller size, no frosting.
- **Why does aircraft use air cycle?** Light weight, air freely available, safe, high pressure bleed air is on board.
- **Ideal vs actual Bell-Coleman?** Actual has irreversible compression/expansion (η_c, η_e), pressure drops and heat leak; COP lower.
- **Define EER.** BTU/h of cooling per watt of electric input; EER = 3.412 × COP.

Likely exam questions

- State and explain Clausius and Kelvin-Planck statements; define COP, EER and TR. (short)
- Derive the COP of reversed Carnot cycle. Why is it not used in practice? (long)
- Explain the Bell-Coleman cycle with p-V and T-s diagrams; derive its COP. (long)
- Compare open and dense air refrigeration systems. (short)

- Explain refrigeration needs of aircraft; compare the types of air-cooling systems. (long)
- Numerical: air-refrigeration cycles with efficiencies; TR and power calculation.